**material_types** 材料类型
id PK
name / stage_code / is_required
title_keywords
format_requirements JSONB

**checklist_items** 阶段材料清单
id PK
stage_code FK → stages
type_id FK → material_types
count_mode / fixed_count

**rule_versions** 规则版本快照
id PK
version_no / snapshot JSONB
change_summary
created_by FK → users

**review_records** 审核留痕
id PK
submission_id FK → submissions
level / action / round_no
reviewer_user_id FK → users

**audit_logs** 审计日志
id PK
actor_user_id FK → users
action / object_type / object_id
触发器禁止改删

**jobs** 后台任务
id PK
job_type / status / ref_id
parent_job_id FK → jobs
attempts / heartbeat_at

**messages** 站内消息
id PK
user_id FK → users
msg_type / title / body
is_read / created_at